SOCIETY FOR COMPUTER TECHNOLOGY & RESEARCH'S

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

AN AUTONOMOUS INSTITUTE AFFILIATED TO SPPU, PUNE

DEPARTMENT OF INFORMATION TECHNOLOGY

40-Hours Value Added Certificate Course on 'Java Backend Developm Boot'

Schedule of Course

Start Date: 16 March 2026 to 2 April 2026

Time: 6 PM to 8 PM

	Session Content (Theory + Lab Assignments)	**Hrs**
1	Inaugration Ceremony	30 min
2	**OOP Foundations: Introduction, OOP principles, Basic Concepts, Allocation/Deallocation, Basic Structure,** Encapsulation, Abstraction. **Lab:** Design Employee class with attributes & methods.	2
3	**Inheritance & Polymorphism. Lab:** Permanent, Contract, Intern employee types with salary calculation logic.	2
4	**Collections Framework:** List, Set, Map, Generics. **Lab:** Store employee records using Map; implement search/update/delete.	2
5	**Exception Handling & Custom Exceptions. Lab:** Create InvalidEmployeeException & validation layer.	2
6	**Multithreading Basics:** Thread class, Runnable, lifecycle. **Lab:** Create booking simulation thread logs.	2
7	**Synchronization & Concurrency. Lab:** Ticket Booking System with synchronized block; demonstrate race condition.	2
8	**Need for Frameworks & Spring Overview.** Discussion: Traditional vs Framework-based backend.	2
9		—
10	**Spring Boot Introduction & Project Structure. Lab:** Create first Spring Boot app & observe startup logs.	2
11	**Starters & Maven Dependency Management. Lab:** Modify pom.xml and analyze dependency errors.	2
12	**Application Configuration & Properties. Lab:** Read external properties; create dev & prod profiles.	2
13	**IoC & Dependency Injection. Lab:** Create Service & Repository with Autowiring.	2

14	**Annotations & Component Interaction. Lab:** Break DI intentionally and debug bean creation error.	2
15	**Logging & Global Exception Handling. Lab:** Implement centralized exception handler.	2
16	**Layered Architecture & Request Flow. Lab:** Create basic REST controller.	2
17	**Sunday (Excluded)**	—
18	**REST API Development (CRUD). Lab:** Student CRUD endpoints using Controller–Service–Repository.	2
19	**Data Persistence Concepts & Entity Mapping. Lab:** Create Student entity & connect with in-memory DB.	2
20	**Spring Data JPA & Repository Pattern. Lab:** Implement CRUD via JpaRepository; add custom finder method.	2
21	**Validation & Testing Overview. Lab:** Add input validation & error response customization.	2
22		—
23	**Mini Project Development – Submission.** Academic Management Backend (Student + Course Module).	4